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Opportunities and Challenges of Utilizing Artificial Intelligence Tools: Faculty Perceptions in Chemistry Department at King Saud University

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Abstract:

This study explores the perceptions of the chemistry teaching staff at King Saud University regarding the opportunities and challenges of using artificial intelligence tools in their teaching methodologies. The descriptive survey approach was used to achieve the study's objectives, and the questionnaire tool was designed, which included two sections and 16 items. The study sample consisted of 49 faculty members. The findings reveal a generally positive perception towards artificial intelligence's role in enhancing educational quality and efficiency. Moreover, faculty members acknowledge artificial intelligence's potential to alleviate educational burdens and boost student engagement. However, concerns persist about artificial intelligence's impact on academic integrity. The study underscores the importance of specialized training for faculty and establishing clear ethical guidelines for artificial intelligence use in higher education. These insights are crucial for developing policies and strategies that align with Saudi Vision 2030.

Keywords: Artificial intelligence tools, opportunities and challenges, chemistry education, chemistry teaching staff.

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فرص وتحديات استخدام أدوات الذكاء الاصطناعي: تصورات أعضاء هيئة التدريس في قسم الكيمياء بجامعة الملك سعود

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المستخلص:

هدفت الدراسة إلى الكشف عن تصورات أعضاء هيئة تدريس قسم الكيمياء بجامعة الملك سعود حول فرص وتحديات استخدام أدوات الذكاء الاصطناعي في طرق تدريسهم لمقررات الكيمياء. وتم استخدام المنهج الوصفي المسحي لتحقيق أهداف الدراسة ، كما تم تصميم أداة الاستبيان التي تضمنت قسمين و16 بنداً. وتكونت عينة الدراسة من 49 عضو هيئة تدريس بقسم الكيمياء بجامعة الملك سعود. وتكشف النتائج عن تصور إيجابي بشكل عام تجاه دور أدوات الذكاء الاصطناعي في تعزيز جودة وكفاءة التعليم. علاوة على ذلك ، يعترف أعضاء هيئة التدريس بإمكانية الذكاء الاصطناعي في تخفيف الأعباء التعليمية وتعزيز مشاركة الطلاب. ومع ذلك ، لا تزال هناك مخاوف بشأن تأثير الذكاء الاصطناعي على النزاهة الأكاديمية. تؤكد الدراسة على أهمية التدريب المتخصص لأعضاء هيئة التدريس ووضع مبادئ توجيهية أخلاقية واضحة لاستخدام الذكاء الاصطناعي في التعليم الجامعي. تعتبر هذه الأفكار ضرورية لتطوير السياسات والاستراتيجيات التي تتوافق مع رؤية السعودية 2030.

الكلمات المفتاحية: أدوات الذكاء الاصطناعي ، الفرص والتحديات ، تعليم الكيمياء ، هيئة تدريس الكيمياء.

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1. Introduction.

Chemistry courses are challenging across educational levels (Treagust et al., 2018). Vincent-Ruz and Boase (2022) noted that abstract concepts, mathematical reasoning, and laboratory skills require significant cognitive effort. Moreover, Hartman et al. (2022) highlighted that chemistry's focus on problem-solving and conceptual understanding makes it one of the most demanding subjects, necessitating dedicated effort and innovative teaching methods for student success.

Students encounter various challenges in chemistry courses due to several factors. One major issue is a lack of understanding of foundational concepts early in their chemistry education, which can cause persistent difficulties (Elhadary & Elhaty, 2021). Scott (2012) highlighted that mathematical components of chemistry, such as molar or stoichiometric calculations, are frequent areas of struggle. Additionally, Salame et al. (2020) noted that conceptual understanding and predicting organic reaction products pose significant challenges, particularly for graduate students. Santos and Mooring (2022) suggested that mindset-related barriers also impact performance, while Asamoah (2024) identified time management, motivation, and course content as additional challenges.

With the considerable advancement of technology, integrating technology and artificial intelligence (AI) tools in teaching chemistry courses has become important in improving students' learning, engagement, and performance (Almasri, 2024). Furthermore, Almasri (2024) stated that advanced technologies such as virtual laboratories and AI-driven tutoring systems provide students with interactive and immersive learning experiences that traditional methods cannot offer.

Rapid progress in AI technology offers promising prospects to revolutionize educational methodologies. Nevertheless, Nja et al. (2023) asserted that there is a lack of comprehensive insight into educators' viewpoints and factors influencing their embrace of AI, posing a significant obstacle to

implementation of these technologies. Thus, exploring teaching staff perceptions to bridge these gaps and facilitate the effective integration of AI tools in science education is necessitated.

With technology's continuous advancement, countries worldwide strive to transition from their traditional models to those based on knowledge and learning. They are utilizing modern technologies which have brought about significant changes in many sectors and transformed the features of our world today (Markauskaite et al., 2022). Elhajji et al., (2020) hinted that due to its impact on many fields, including education at all levels, education must keep pace with this progress and development, improving the use of artificial intelligence in achieving the goals related to Saudi Vision 2030 (Saudi Vision 2030, 2016).

The main objective of this research is to examine the viewpoints of teaching faculty members at King Saud University on the application of artificial intelligence (AI) tools in teaching science subjects to improve the integration of AI tools and the quality of science education in Saudi Arabia. Specifically, this investigation seeks to understand the general perception of teaching staff towards implementing AI tools in science courses and identify the challenges they face in utilizing AI technology in these classes.

2. Review of the Literature

Artificial intelligence (AI) tools have attracted considerable interest among teaching faculty in delivering science courses (e.g., chemistry) within university settings. This study explores the perceptions of chemistry teaching staff at King Saud University regarding the opportunities and challenges of AI tools methodologies.

2.1. Constructivism and Technology in Education

Constructivism is a learning theory suggesting that individuals create their own understanding and knowledge by engaging in experiences and

reflecting on those experiences. This theory highlights the significance of a learner's background, culture, and language in the learning process. It views learning as a social activity, with knowledge constructed through interactions and collaborations (Fosnot, 2013). Moreover, according to Bada and Olusegun (2015), constructivism emphasizes that students actively construct their own knowledge through meaningful experiences, making learning more student-centered, interactive, and practical. This approach contrasts with traditional methods and has significant implications for modern educational practices.

Technology has become central to education, with constructivism being a key theory discussing its use. Mota-Valtierra et al. (2019) noted that constructivist approaches in AI education enhance skills by building on prior knowledge and promoting analytical thinking. Similarly, Grubaugh et al. (2023) stated that constructivism's focus on experiential learning aligns with AI's adaptive capabilities, offering tailored and interactive educational experiences for diverse learners. Constructivism asserts that learners build understanding through active engagement rather than passive absorption (Grubaugh et al., 2023).

Tuomi (2024) suggested that some forms of constructivism stress the social aspects of knowledge acquisition, while others focus on the technological and material contexts shaping cognition. However, all forms view learning as dynamic and innovative, surpassing the limitations of operant conditioning. Additionally, Tuomi highlighted that cultural-historical activity theory uniquely regards the material environment as integral to cognition and learning processes. Vygotsky noted that knowledge is embedded in culturally accumulated pieces, with learning rooted in tangible, "object-oriented" praxis rather than abstract representations or isolated ideas.

The shift to phenomenological epistemology reframes "knowing" as a process rather than static "knowledge," challenging traditional cognitive and AI learning models (Tuomi, 2024). Thinkers like Whitehead and Bergson

thus gain relevance in AI research , prompting a reassessment of foundational concepts. Zhai et al. (2021) argued that AI's limited impact on teaching stems from inadequate pedagogical frameworks and assessment metrics, emphasizing the need for future studies grounded in educational theories to enhance AI's role in learning.

2.2. Benefits of Using AI in Education

AI tools have proven highly effective in enhancing educational outcomes in science education , including chemistry. For instance , ChatGPT has been shown to improve critical thinking, problem-solving, conceptual understanding, creativity, and personalized learning in chemistry courses, based on data from interaction logs and reflective journals (Santos, 2023). Similarly, Harry and Sayudin (2023) emphasized AI's role in fostering personalized learning and providing timely, actionable feedback. AI's adaptive functionalities enable the creation of customized course materials, practice quizzes, and exam queries, tailoring instruction to learners' backgrounds and prior knowledge (Walkington, 2013). Furthermore, these technologies aim to provide tailored, personalized learning experiences (Sharma et al., 2017) while fostering advanced skills like statistical programming and data analysis (Ellis & Slade, 2023).

AI technologies also have the potential to introduce global perspectives into education, enriching research and project outcomes. A study of fifty-three teachers revealed that AI tools enable students to simulate real-life scenarios, creating a more engaging and hands-on learning environment (Antonenko & Abramowitz, 2023). Additionally, Makhmudovich (2023) highlighted that AI in classrooms allows faculty to focus on fostering students' creativity and innovation while significantly reducing time spent on repetitive tasks. Furthermore, automated grading systems help educators prioritize teaching content by significantly decreasing the time required for evaluating exams and assignments (Zaman, 2024).

2.3. Issues in Using AI Tools in Education

Despite its potential, university teaching staff have raised concerns about integrating AI into science courses (Schroeder, 2023). Alasadi and Baiz (2023) emphasized the need to address these challenges, particularly in teaching chemistry. A primary concern is academic integrity, as the ease with which students use AI tools to generate code demands a reevaluation of assignment design (Hocky & White, 2022). Additionally, the use of AI tools like ChatGPT for essay writing raises issues about the authenticity of student work and the development of critical writing skills (Humphry & Fuller, 2023).

Harry and Sayudin (2023) highlighted concerns associated with AI, including issues of privacy, trust in AI-generated information, and the cost of implementation. Cooper (2023) added concerns about content moderation, copyright infringement, and the need for explicit guidelines for AI use in evaluations. Moreover, Su and Yang (2023) noted that the effectiveness of AI and the quality of its outputs remain untested, and AI tools often struggle with complex tasks, emphasizing the need for a balanced approach to AI integration.

In an experiment involving two users testing ChatGPT's responses to introductory chemistry course questions, it was found that current AI models face limitations in addressing specific queries effectively (Leon & Vidhani, 2023). These constraints necessitate careful monitoring to ensure accuracy and relevance in AI-generated data. Additionally, limitations in AI training datasets raise fairness concerns, particularly given AI's expanding role in education (Emenike & Emenike, 2023).

2.4. Integration of AI Tools in Science Education

The successful integration of artificial intelligence (AI) tools into science education necessitates comprehensive preparation for faculty and staff. Marchak et al. (2021) recommended designing a specialized teacher training program to help educators incorporate AI into chemistry teaching in creative and engaging ways. This preparation ensures faculty can proficiently use AI to enhance

educational outcomes. Faculty members are integral in guiding students to use AI tools , such as chatbots , to articulate ideas and construct coherent arguments , thereby fostering deeper subject understanding (Santos , 2023).

Moreover , faculty are encouraged to model responsible AI use by emphasising critical thinking and clearly communicating guidelines for AI applications in assessments (Cooper , 2023). This approach promotes an ethical and thoughtful integration of technology into academic pursuits. Su and Yang (2023) advised educators to follow a structured process for effective AI use , including defining outcomes , identifying automation levels , addressing ethical concerns , and evaluating AI tool efficacy. Furthermore , Chan (2023) , based on a survey of 457 students and 180 faculty members , highlighted the necessity for universities to establish educational policies governing AI use. Such policies would uphold principles of academic integrity and deter unethical practices like cheating.

2.5. Particular Applications

Artificial intelligence (AI) tools have a range of practical uses in chemistry education. These applications include creating assessment items , class educational materials , email responses , and grading criteria. By automating these administrative functions , educators can allocate more time engaging in interactive teaching methods (Emenike & Emenike , 2023).

2.6. Future Prospects

Faculty teaching members widely recognize the significance of introducing students to AI , given its increasing relevance in various professional fields (Antonenko & Abramowitz , 2023). As AI technologies advance , faculty members must continuously adjust their teaching methods to stay abreast of these technological developments (Leon & Vidhani , 2023). This ongoing transformation presents obstacles and prospects for enriching the educational process and equipping students for a future progressively influenced by AI innovations.

2.7. AI in the Saudi Educational Context

Saudi Arabia has prioritized AI integration into education under its Saudi Vision 2030 initiative to enhance educational quality and diversify the economy. Mutambik (2024) found, in a study of 1991 students, that AI is essential for educational success. Faisal (2024) highlighted that combining innovative theories with practical strategies can help Saudi universities maximize ChatGPT's potential. Alammari (2024), through surveys with 125 educators and interviews with three experts, demonstrated that AI improves student cooperation, personalized learning, and academic achievement. However, Alotaibi and Alshehri (2023) noted that AI use in science disciplines like chemistry remains limited due to staff awareness and acceptance issues, requiring targeted solutions for effective implementation.

3. Research Problem:

The rapid progress in AI technology offers promising prospects for revolutionizing educational methodologies. Moreover, integrating AI in education offers opportunities such as improving learning outcomes, enhancing personalized learning, and preparing learners for an AI-driven world. However, it also presents challenges including ensuring academic integrity, addressing policy implications, and managing the risks associated with AI integration (Pedro et al., 2019).

Furthermore, in response to the recommendations of global and regional conferences on artificial intelligence, including the third Arab Media Conference on "Media in the Age of Artificial Intelligence: Opportunities and Challenges" held in Tunisia on January 17-18, 2024, where the most prominent recommendations were: including artificial intelligence education in all stages of education and training the workforce in artificial intelligence skills (Arab states broadcasting union, 2024).

In addition, Alotaibi and Alshehri (2023) noted a growing interest in using artificial intelligence (AI) in education. However, educators have

raised concerns about issues like integrity , plagiarism , and ethics (Alasadi & Baiz, 2023). Despite the opportunities AI offers, detailed research on its specific challenges and benefits remains limited , particularly regarding higher education outcomes in Saudi Arabia , even as the country advances its national development goals and prioritizes AI integration across sectors.

The integration of AI-based learning in education remains in its early stages. Faisal (2024) highlighted a research gap in understanding the specific opportunities AI could bring to Saudi Arabian higher education. AlKanaan (2022) further noted that AI use in teaching , especially in science education , is limited and inconsistent in Saudi universities due to varying levels of awareness , readiness , and acceptance among educators. This inconsistency is often attributed to a lack of comprehensive understanding of educators' perspectives and the factors influencing their adoption of AI in the classroom.

Therefore , this poses a significant barrier to the widespread adoption of AI technologies , highlighting the need to explore teaching staff perceptions to address these gaps and enable effective integration of AI tools in science education. Consequently , understanding faculty members' views on the opportunities and challenges of using AI in university education becomes essential.

4. Research Questions:

This study examines the perceptions of chemistry teaching staff at King Saud University regarding the opportunities and challenges associated with integrating AI tools into their teaching practices. Additionally , it aims to offer insights that can guide decision-making and policy development to enhance the integration of AI tools and the overall quality of science education in Saudi Arabia. To achieve these objectives , the research addresses the following two questions:

1. What opportunities do the academic staff members at King Saud University see in implementing AI tools in teaching chemistry?

2. What challenges do academic staff members face in utilizing AI technology in chemistry courses?

5. Method and Procedures.

This section addresses the methodological procedures undertaken by the researcher to answer the study's questions.

5.1. Study Methodology.

The descriptive survey method was used in light of the study's nature. This method relies on studying the phenomenon as it exists in reality (Cohen et al. , 2005).

5.2. Study Sample.

Due to the small study population , the study included all faculty members in the Department of Chemistry at King Saud University, totalling 56 members. Of these , 49 faculty members responded , representing 87.5% of the study population.

5.3. Study Tool.

Based on the study's objectives, and the literature review studies employed various methodologies to gather insights. Santos (2023) used interaction logs and reflective journals; Antonenko and Abramowitz (2023) conducted surveys with 53 teachers; Leon and Vidhani (2023) utilized an experimental design with two participants; Chan (2023) carried out a large-scale survey with 457 students and 180 teaching staff; Mutambik (2024) adopted a case study approach analyzing the experiences of 1991 students; and Alammari (2024) employed a mixed-methods design combining surveys of 125 educators with interviews of three expert practitioners. These methodologies underscore the reliability of surveys and other instruments in academic investigations. However, the researcher's instrument is more focused and specific to the teaching staff at one university

department, while the other studies employed a range of methodologies across various contexts to gather broader insights. Therefore, the researcher's utilization of a survey with 56 chemistry staff members at King Saud University is validated by prior research, which underscores the dependability and effectiveness of surveys in academic investigations. Therefore, the researcher found that the appropriate tool is the "questionnaire". So, the study relies on a questionnaire as the primary tool for data collection. Thus, this tool was built and designed by referring to educational literature related to the subject of the study. In its final form, the questionnaire consisted of 16 items divided into two sections: the first section addresses the opportunities of using artificial intelligence tools. This section includes eight items, and the second section addresses the challenges of using artificial intelligence and includes eight items. The researcher adopted a five-point scale according to the following categories:

Table (1)
Determination of the Five-Point Graduated Scale Categories

Strongly disagree	Disagree	Neutral	Agree	Strongly agree
1 - 1.80	1.81 - 2.60	2.61 - 3.40	3.41-4.20	4.21 - 5.0

5.4. Validity of the Study Tool.

The researcher verified the validity of the study tool through the following:

5.4.1 Face validity of the Study Tool.

Cohen et al. (2005, p. 132) highlight that "face validity (that, superficially, the test appears at face value — to test what it is designed to test)." After developing the initial study tool, it was reviewed by five Chemistry faculty members at King Saud University for feedback on its adequacy and potential adjustments. Based on their input, necessary modifications were made, including revising and removing items. The finalized questionnaire, consisting of 16 items, was then distributed electronically to the study participants.

5.4.2 Internal validity.

It was applied in the field after confirming the study tool's face validity. The Pearson correlation coefficient was calculated to determine the questionnaire's internal validity. The correlation coefficient was calculated between the score of each statement in the questionnaire and the total score of the section to which the statement belongs, as shown in the following tables.

Table (2)

Pearson correlation coefficients for the items on the sections of the study tool with the total score for each section

The first section		The second section	
Item	Correlation Coefficient	Item	Correlation Coefficient
1	0.62	9	0.57
2	0.64	10	0.62
3	0.59	11	0.53
4	0.66	12	0.65
5	0.66	13	0.54
6	0.70	14	0.56
7	0.55	15	0.51
8	0.60	16	0.63

The results in Table 2 indicate that all correlation coefficients are significant at the 0.01 level, with values ranging from 0.51 to 0.70. This demonstrates high internal validity and suggests that the study tool possesses acceptable and sufficient validity indicators for reliable application.

5.5. Reliability of the Study Tool.

The researcher used Cronbach's Alpha to measure the reliability of the study tool. Table (3) shows the reliability coefficients for each category of items in the study tool. The questionnaire items are classified into five categories to reflect the main aspects under which they are classified.

The categories are:

1. Enhancing Learning and Engagement.
2. Reducing Educational Burden.
3. Improving Education Quality.
4. Challenges and Concerns.
5. Reliability and Training.

Table (3)

Cronbach's alpha coefficient for measuring the reliability of the study tool

No.	Categories	Reliability Coefficient
1	Enhancing Learning and Engagement	0.85
2	Reducing Educational Burden	0.78
3	Improving Education Quality	0.82
4	Challenges and Concerns	0.75
5	Reliability and Training	0.80
Overall Reliability		0.80

Table (3) shows that the study questionnaire has statistically acceptable reliability. The total reliability coefficient (alpha) value reached (0.80) , an excellent reliability score. The reliability coefficients of the study tool ranged from (0.75 to 0.85) , which are high-reliability coefficients that can be trusted in applying the current study tool.

The reliability coefficients for the study tool ranged from 0.75 to 0.85 , indicating high reliability overall. The researcher concludes these scores are sufficiently robust to support the tool's application. The "Improving Learning and Engagement" section achieved the highest reliability coefficient of 0.85 , reflecting strong reliability , while "Challenges and Concerns" had the lowest at 0.75 , suggesting potential areas for refinement. The mean reliability coefficient across sections was approximately 0.80 , with a standard deviation of 0.034 , demonstrating consistent reliability and a moderate spread between sections.

6. The results of the study:

The following part presents the results of the questionnaire:

6.1. The first section is the opportunities for using artificial intelligence tools.

Table (4)

Cronbach's alpha coefficient for measuring the reliability of the study tool

No.	Item	Mean	Standard Deviation	Percentages				
				Strongly agree	Agree	Neutral	Disagree	Strongly disagree
1	Artificial intelligence enhances students' self-learning.	3.816	0.858	18.37%	55.1%	16.33%	10.20%	0%
2	Artificial intelligence contributes to raising students' participation and engagement.	3.857	0.866	22.45%	48.98%	20.41%	8.16%	0%
3	Integrating artificial intelligence tools in education is vital for improving education quality.	4.041	0.889	26.53%	61.22%	6.12%	2.04%	4.08%
4	Artificial intelligence tools reduce the educational burden on faculty members.	3.531	1.002	16.33%	38.78%	28.57%	14.29%	2.04%
5	Artificial intelligence tools save time for faculty members and students.	3.959	0.706	16.33%	67.35%	14.29%	0%	2.04%
6	Artificial intelligence tools develop students' critical thinking skills.	3.449	0.738	2.04%	53.06%	32.65%	12.24%	0%
7	Artificial intelligence tools consider individual differences among students.	3.0	0.913	4.08%	26.53%	36.73%	30.61%	2.04%

No.	Item	Mean	Standard Deviation	Percentages				
				Strongly agree	Agree	Neutral	Disagree	Strongly disagree
8	Artificial intelligence tools enhance students' cooperation and communication within the classroom.	3.429	0.842	4.08%	48.98%	36.73%	6.12%	4.08%

The assessment of faculty members' perspectives on AI tools in education reveals a generally optimistic outlook, particularly regarding improving educational quality, fostering self-directed learning, and enhancing student engagement. Faculty members strongly agreed on the potential of AI to enrich learning outcomes, reflected in the highest mean score of 4.041, indicating widespread support.

There was also agreement on AI's timesaving benefits for both faculty and students, with a mean score of 3.959 and a low standard deviation of 0.706, highlighting a shared recognition of AI's efficiency. Additionally, AI was seen as a significant aid in promoting self-learning (mean = 3.816) and increasing student participation in educational activities (mean = 3.857). However, moderate standard deviations (0.858 and 0.866, respectively) suggest some variability in enthusiasm, indicating that while the majority value AI's benefits, individual perspectives vary.

In contrast, faculty opinions on the role of AI in alleviating the academic workload of staff members and nurturing students' critical thinking abilities are more balanced. The mean score for AI's contribution to reducing the educational burden stands at 3.531, accompanied by a higher standard deviation of 1.002, indicating a diversity of opinions. The results indicate a generally favourable view of AI's contribution to enhancing the quality and efficiency of education, with notable acknowledgement of its ability to lessen educational burdens. Similarly, the mean score of 3.449 for AI's role in enhancing critical thinking skills reflects a balanced held belief, with a standard deviation of 0.738, showing reasonable agreement and some uncertainty.

Regarding AI's capacity to consider individual differences among students , the neutral mean score of 3.0 implies a balanced stance , with a higher standard deviation of 0.913 pointing towards a wide array of opinions. This underscores a potential area of concern , as faculty staff members exhibit divergent views on whether AI effectively addresses the varied needs of students. Lastly , the mean score of 3.429 for AI enhancement of cooperation and communication within the classroom portrays a moderately positive perspective , with moderate consensus (SD = 0.842) among the faculty staff members.

The analysis of each statement according to the Likert scale for this section will be represented as follows: each inquiry is assessed according to the percentage of responses indicating participants' levels of consensus or dissent.

For enhancing students' learning , the findings reveal that a considerable majority , 73.47% (Strongly agree + Agree) , perceive that artificial intelligence bolsters self-directed learning. A lesser segment maintains a neutral stance (16.33%) , whereas a negligible portion (10.2%) expresses disagreement. The lack of pronounced dissent indicates a prevailing favourable agreement regarding the function of AI in facilitating self-learning. Regarding participation and engagement , 71.43% of participants concur or strongly concur with the assertion , indicating a widespread conviction in AI's capacity to augment engagement. The proportion of neutral responses is marginally more significant (20.41%) than that observed in the initial inquiry. In contrast , the percentage of dissent remains minimal (8.16%) , reinforcing AI's favourable view in promoting engagement.

As for education quality , 87.75% of participants concur or strongly concur that artificial intelligence is essential for improving educational quality. The negligible proportions of neutral (6.12%) and dissenting (6.12% collectively for Disagree and Strongly Disagree) responses signify a robust consensus supporting the incorporation of AI within the educational framework. The proportion of consensus (55.11%) indicates that a predominant segment perceives artificial intelligence as a means to alleviate

educational challenges; however, a significant 28.57% maintain a neutral stance, while 16.33% either dissent or express strong disapproval, reflecting certain hesitations or acknowledged constraints.

A significant proportion (83.68%) of respondents concur or strongly concur that artificial intelligence facilitates timesaving, with a modest neutral response (14.29%) and minimal dissent (2.04%). This indicates a robust affirmation of AI's efficacy within educational contexts. Although 55.1% concur that artificial intelligence fosters critical thinking abilities, the substantial proportion of neutral responses (32.65%) coupled with a moderate degree of dissent (12.24%) indicates divergent perspectives regarding the efficacy of AI in this domain. The data revealed in this analysis indicate a split perspective, with only 30.61% in alignment, while a considerable proportion remains neutral (36.73%) or in disagreement (32.65%). This implies that the efficacy and clarity of AI's function in accommodating individual variances are regarded as ambiguous or inadequate. Finally, 53.06% agree that AI enhances cooperation and communication, though 36.73% remain neutral. The remaining 10.20% disagree, reflecting some uncertainty or scepticism about AI's impact in this area.

In response to the first research question about the opportunities of implementing AI tools in teaching Chemistry from the side of the academic staff members, it can be concluded that there is a positive perception. However, while there is an acceptable recognition and a positive perception of the opportunities of AI tools in improving educational quality, self-learning, participation, and efficiency, there are some reservations regarding the statement "Artificial intelligence tools enhance students' cooperation and communication within the classroom." These findings indicate that although AI is generally perceived as a valuable asset in education, further efforts are imperative to address specific apprehensions and ensure the effective integration of AI tools into educational settings to maximize their potential benefits.

6.2. The second section: the challenges of using artificial intelligence.

Table (5)

Means, Standard Deviations and Percentages of the challenges of utilizing AI tools

No.	Item	Mean	Standard Devia- tion	Percentages				
				Strongly agree	Agree	Neutral	Disagree	Strongly disagree
9	Artificial intelligence tools threaten academic integrity	3.571	0.913	14.29%	42.86%	28.57%	14.29%	0%
10	Students' reliance on artificial intelligence tools reduces their reference to sources and scientific research	4.143	0.890	38.78%	44.9%	8.16 %	8.16%	0%
11	Regulations and guidelines are available for students' use of artificial intelligence tools in education	2.327	1.029	2.04 %	10.20%	30.61%	32.65%	24.49%
12	Using artificial intelligence requires high skills to achieve the desired results	3.673	1.068	20.41%	48.98%	10.20%	18.37%	2.04%
13	Using artificial intelligence requires much effort from faculty members and students	3.224	0.985	6.12%	42.86%	18.37%	32.65%	0%
14	Faculty and students use of artificial intelligence tools is financially costly	3.143	1.000	6.12%	36.73%	24.49%	30.61%	2.04%
15	The information and data obtained by faculty and students from artificial intelligence tools are reliable	2.653	0.879	0%	18.37%	36.73%	36.73%	8.16%
16	Faculty members and students at the university have specialized training programs in artificial intelligence and its tools	2.857	1.099	4.08%	26.53%	34.69%	20.41%	14.29%

Data analysis of faculty members' perspectives on potential challenges and requirements associated with using artificial intelligence (AI) tools in the educational sector brings to light numerous apprehensions. Among the notable issues identified is the concern that AI tools threaten academic integrity, as evidenced by a mean score of 3.571 and a standard deviation of 0.913. This statistical data signifies that many faculty members have reservations about the possibility of AI being misapplied, which could compromise the genuineness of academic endeavours. Furthermore, this apprehension is exacerbated by the belief that students' dependence on AI tools diminishes their interaction with regular sources and scientific investigations, as indicated by the highest mean score of 4.143 and a standard deviation of 0.890. These findings suggest a strong consensus within the academic community on the potential detrimental impact of AI on students' critical engagement with scholarly materials, raising concerns about the enduring consequences of academic rigour.

In addition, the perceived inadequacy of the regulations and guidelines governing the use of AI in educational settings is a prevalent concern (Erdélyi & Goldsmith, 2018), with a low mean score of 2.327 and a relatively high standard deviation of 1.029. This variability in responses implies that, while certain institutions have established regulatory frameworks, others have not, resulting in inconsistencies in the management of AI in educational settings. Similarly, the notion that proficiency in using AI tools requires high levels of expertise to achieve desired results (Poldrack et al., 2023) (mean = 3.673, SD = 1.068) underscores the intricate nature of AI tools and the potential hurdles they present for both students and faculty in mastering their functionalities. Therefore, Yuan et al. (2024) claimed a need to spread AI literacy and proficiency.

The perspective that integrating AI requires substantial dedication from faculty members and students is corroborated by a mean score of 3.224 and a standard deviation of 0.985, indicating a moderate consensus on the challenges of integrating AI into education. Additionally, the financial implications of AI

tools are viewed as a pressing issue , with a mean score of 3.143 and a standard deviation of 1.000, underscoring the significant economic burden of implementing AI in educational settings for many faculty staff members.

Another focal point of concern pertains to the statement, “The information and data obtained by faculty and students from artificial intelligence tools are reliable”, with a mean score of 2.653 and a standard deviation of 0.879. This relatively low mean value reflects scepticism regarding the accuracy and reliability of AI-generated content , potentially eroding faculty and students' trust in these tools. Finally, it is noted that there is a deficiency in specialized AI training programs in universities, which is evident from a mean score of 2.857 and a standard deviation of 1.099. While some academic institutions may offer such programs , a substantial need exists for more extensive and inclusive training initiatives to ensure that faculty and students are adequately equipped to use AI tools proficiently.

The analysis of each statement according to the Likert scale for this section will be represented as follows: each inquiry is assessed according to the percentage of responses indicating participants' levels of consensus or dissent.

The findings indicate that 57.15% (Strongly agree + Agree) of participants perceive AI tools as a potential risk to academic integrity. A considerable fraction maintains a neutral stance (28.57%), whereas an identical 14.29% express disagreement, suggesting a polarised viewpoint accompanied by significant apprehension regarding AI's influence on academic honesty. A substantial fraction , 83.68%, affirm or strongly affirm that reliance on artificial intelligence detrimentally influences students' interaction with conventional research materials. A mere minority , 8.16%, remains ambivalent or disagrees , suggesting a robust agreement regarding the potential detrimental influence of AI on research practices.

For academic integrity , the findings reveal a significant apprehension about regulations and guidelines , as evidenced by 57.14% of participants expressing disagreement or strong disagreement. Conversely, a mere 12.24% affirm the presence of such regulations , while a notable 30.61% adopt a neutral stance ,

underscoring the perceived absence of clearly defined guidelines in this domain. An overwhelming majority of participants (69.39%) acknowledge or strongly acknowledge that skilled employment of AI entails advanced skills. In contrast, a significant minority demonstrates disagreement (18.37%) or neutrality (10.20%), suggesting a divergence in views about the required skill level.

The data demonstrates a split perspective regarding the effort associated with AI deployment. 48.98% agree or strongly agree, contrasting with 32.65% who disagree. This underscores acknowledging the required effort tempered by significant opposition, potentially driven by divergent anticipations. As for the cost, the findings reveal a balanced distribution of perspectives, with 42.85% concurring on the high costs of AI tools, 32.65% dissenting, and a notable 24.49% remaining neutral, underscoring the varied opinions regarding the economic implications of AI implementation.

The responses reveal considerable scepticism regarding the credibility of AI-generated information. 44.89% of respondents express disagreement, 36.73% remain neutral, and 18.37% endorse its reliability, indicating apprehensions about AI's trustworthiness in scholarly settings. Regarding training programs, the data here reveals a divided perception regarding the accessibility of AI training, evidenced by 30.61% of respondents expressing agreement, contrasted with a greater proportion of 34.70% indicating disagreement. A significant neutral segment of 34.69% suggests ambiguity in the availability of specialized training.

In conclusion, the analysis reveals, in answering the second research question about the challenges that face academic staff members in utilizing AI technology in chemistry courses, that while artificial intelligence (AI) tools offer potential benefits, significant concerns regarding their impact on academic integrity, implementation challenges, and financial implications persist. Additionally, there is a notable lack of sufficient regulations, reliable information sources, and specialized training for faculty and students in the use of AI tools, emphasizing the need for ethical considerations in their educational application (Bibi et al., 2024)

7. Discussion

This section aims to discuss the findings of the study related to the opportunities and challenges of utilizing AI tools , and this is because this study has evaluated the perception of the chemistry department teaching staff at King Saud University.

7.1. Perceptions towards AI in Science Education

The literature review emphasized the increasing interest in using artificial intelligence tools in science education (particularly chemistry) , acknowledging the potential opportunities and challenges (Faisal, 2024). This research supports these conclusions, showing that faculty staff members generally recognize the capability of AI to enhance learning effectiveness. Incorporating AI tools such as ChatGPT has been proven to enhance critical thinking, problem-solving abilities, and personalized learning experiences (Santos, 2023). This corresponds to the results, where survey participants expressed a favourable perception towards AI's capacity to promote self-directed learning and increase student engagement (mean = 3.82, SD = 0.86). It is crucial to embrace continued advancement, ongoing adaptation, and vigilance to fully leverage the potential of AI while addressing its constraints and ensuring fair access to its opportunities.

7.2. Opportunities for AI Integration

The literature review has attracted attention to the increasing interest in utilizing AI tools in research and academic settings. Both the study and the literature emphasize the considerable benefits that AI tools provide (Faisal, 2024). The adaptive learning capabilities of AI make it possible to generate personalized educational materials, thus catering to various student needs and promoting inclusivity in educational settings (Ellis & Slade, 2023). The faculty members and the research participants acknowledged these opportunities, especially in reducing the academic workload and allowing them to concentrate more on interactive teaching methods (mean = 3.96,

SD = 0.71). Moreover, the role of AI in broadening students' global perspectives and facilitating hands-on learning environments was highlighted by Antonenko and Abramowitz (2023).

7.3. Challenges and Concerns

Despite the positive elements, significant issues surround incorporating artificial intelligence (AI) into education. Concerns such as maintaining academic integrity, overseeing content, and ensuring the accuracy of AI-generated data were highlighted in various academic works (Hocky & White, 2022; Humphry & Fuller, 2023). The research results align with these concerns, showing an apparent unease among faculty members about the potential impact of AI tools on academic integrity and the possible reduction in students' dependence on typical and common research sources such as books and scientific references, academic articles (mean = 4.14, SD = 0.89). The need for comprehensive guidelines and ethical frameworks to govern the use of AI in educational settings is evident.

7.4. Implementation and training

Coopers (2023) literature review suggests that effective AI integration requires thorough training of faculty staff members and establishing clear guidelines. The research findings align with this notion, as the survey respondents expressed the importance of specialized training initiatives to improve their proficiency in AI tools (mean = 2.86, SD = 1.10). Faculty staff must possess the necessary skills to navigate AI's ethical considerations and pedagogical uses to optimize its opportunities while minimizing potential risks.

7.5. Specific Applications in Chemistry Education

In chemistry education, artificial intelligence (AI) tools have been used for various purposes, such as creating assessments, developing content, and handling administrative responsibilities (Emenike & Emenike, 2023). The research results have validated the use of these functions, as educators

have acknowledged the efficacy of AI in reducing repetitive tasks, thus enabling more meaningful educational engagements. Furthermore, Santos (2023) indicated that AI enhances chemistry education by offering personalized learning, real-time tutoring, automated assessments, content creation, and collaborative tools, making learning more personalized, interactive, and efficient. Additionally, AI tools can assist teachers in chemistry education through intelligent tutoring systems, adaptive platforms, and enhanced accessibility to student learning materials (Iyamuremye et al., 2024). Also, Alasadi and Baiz (2023) added that the applications of AI in chemistry education include virtual laboratories and simulations, enhanced accessibility for students, and advanced research and data analysis. Moreover, Emenike and Emenike (2023) underscore the advantages of AI tools like ChatGPT in enhancing chemistry education by aiding in writing tasks, creating assessment questions, elucidating complex concepts, and supporting literature reviews. However, they also emphasize the importance of addressing issues related to academic integrity, content accuracy, and establishing ethical guidelines. Finally, Vincent-Ruz and Boase (2022) emphasize the benefits of AI-driven adaptive learning in chemistry education, such as fostering discipline-specific thinking, boosting engagement and retention, ensuring equitable learning opportunities, and promoting effective educational behaviours.

However, concerns have been raised about the inability of current AI models to adequately address intricate inquiries, particularly in foundational courses (Leon & Vidhani, 2023).

7.6. Future Prospects and Recommendations

Looking forward, the potential expansion of AI's role in teaching chemistry, specifically at the university level, will necessitate continual adaptation from educators (Antonenko & Abramowitz, 2023). The research proposes several suggestions for future implementation and policy.

1. Thorough training: Establish vital training initiatives for chemistry department faculty members to improve their proficiency in using AI tools and guiding students in their ethical application.
2. Ethical Guidelines: Develop explicit, organization-wide policies and ethical frameworks to regulate the use of AI in teaching chemistry environments.
3. Allocate resources towards AI infrastructure and support systems to facilitate smooth integration and efficient utilization in educational settings for chemistry teaching.

8. Conclusion.

This study highlights the transformative potential of AI tools in chemistry education, as perceived by faculty members at King Saud University. They recognize AI's ability to enhance educational quality, promote self-directed learning, and boost student engagement. Faculty members also appreciate AI's saving time and reducing instructional burdens, allowing for more interactive and innovative teaching methods. However, there are significant challenges, including concerns about academic integrity, students' over-reliance on AI at the expense of traditional research practices, and doubts about the reliability of AI-generated information. The lack of comprehensive regulations, ethical guidelines, and specialized training programs hinders effective AI integration.

These insights emphasize the need for a balanced approach to AI implementation in higher education. While AI offers substantial benefits, its adoption should be accompanied by ethical oversight, clear regulatory frameworks, and targeted training initiatives to address its challenges and maximize its potential. Further research should explore diverse educational contexts and evaluate the long-term impacts of AI integration to deepen understanding and inform policy development.

The study faces some limitations that may affect the generalizability and applicability of its findings. Firstly, it was limited to a single department at one institution, which restricts the broader applicability of the results to other academic contexts or disciplines. Additionally, the reliance on self-reported perceptions from faculty members may not fully capture AI tools' practical effectiveness or challenges in real-world teaching scenarios. Finally, the absence of longitudinal data further limits the ability to assess the long-term impact of AI integration on educational outcomes, leaving a gap in understanding its sustained effectiveness. Future research should address these limitations to understand AI's role in teaching chemistry comprehensively.

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